Laboratory Data Modelling

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Laboratory Data Modelling (A06220)

Short Title:	Laboratory Data Modelling		
Faculty	Science and Computing		
Department	Science		
Cluster	IT		
Author	KGRENNAN		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

In this module the student will develop skills to fit linear and non-linear theoretical models to experimental data. This will enable the student to quantitatively analyse a spreadsheet containing a set of data, fit appropriate models to these and evaluate the fit of such models. This module will also build on the fundamentals of descriptive and inferential statistics to analyse laboratory data. These techniques will be used to analyse laboratory data using relevant statistical software.

Programmes

SCIE-0038	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	3 / 6 / M
SCIE-0038	BSc in Pharmaceutical Science (WD_SPHSC_D)	3 / 6 / M

Indicative Content

- Use of MS Excel's Data Analysis ToolPak to perform linear regression analysis on laboratory data
- Use of MS Excel's Solver program to perform non-linear regression analysis on laboratory data
- Techniques to evaluate the quality of fit of linear & non-linear trendlines to data
- Analysis of the Normal, Binomial and Poisson distributions with respect to analysing laboratory data
- Inferential statistical testing, including F-tests and t-tests for the validation of laboratory results
- Design of experiments and ANOVA models including one-way ANOVA and two-way ANOVA

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Apply linear & non-linear regression analysis to experimental data sets.
2. Examine the fit of linear & non-linear regression models to experimental data.
3. Develop a critical attitude to experimental data analysis and modelling laboratory data.
4. Apply and interpret a range of statistical tests to laboratory data.
5. Apply and interpret appropriate ANOVA models to laboratory data.
6. Employ a variety of techniques on statistical software packages to analyse laboratory data.

Learning and Teaching Methods

- Lectures.
- IT practicals.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
In-Class Assessment	35%	5,6
Assignment	15%	4
Assignment	50%	1,2,3

Assessment Criteria

<40%: No demonstration of an understanding of the fundamentals of the learning outcomes, selection of the correct mathematical models to fit to laboratory data and understanding of statistical techniques.

40%-49%: Will be able to show a basic knowledge of the key learning outcomes including demonstration of rudimentary modelling skills with respect to fitting appropriate mathematical models and applying statistical tests to data.

50%-59%: As well as achieving the pass standard, the student will be able to demonstrate an understanding of some of the complexities of the topics and demonstrate fundamental modelling skills with respect to fitting mathematical models and applying statistical tests to laboratory data.

60%-69%: In addition, the student will demonstrate more detailed knowledge of the all topics and demonstrate advanced modelling skills with respect to fitting appropriate mathematical models and applying statistical tests to laboratory data.

70%-100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, demonstrate superior modelling skills with respect to fitting mathematical models and applying statistical tests to laboratory data.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	18	
Practical	18	
Independent Learning	99	

Essential Material

- "Laboratory Data Modelling module Moodle page." http://moodle.wit.ie

Supplementary Material

- Montgomery, D. _Introduction to Statistical Quality Control_. 7th ed. USA: Wiley, 2012.

Requested Resources

- Room Type: Computer Lab